

Research Interests and Vision

My research focuses on making robots capable of solving a wide range of tasks without compromising on their performance and reliability. I develop algorithms that learn general models for perception and interaction. Combined with control and planning, these models allow robots to accurately manipulate their environment and meet diverse task requirements.

Work Experience

- 2022-Present **Google DeepMind**. Senior Research Scientist. My main research focus is on dexterous manipulation. To that end, my work includes research on multi-fingered hands, diverse embodiments, visuo-language models, reinforcement learning, behavioral cloning and sim-to-real transfer.
- Summer 2016 **Google Zurich**. Software Engineer Intern at the SafeSearch team.
Research and implementation of deep learning algorithms for a large-scale computer vision problem.

Education

- 2016–2022 **Massachusetts Institute of Technology (MIT)**, *Cambridge*, United States.
- PhD student at the Mechanical Engineering department of MIT. GPA: 5.0.
 - Advisor: Prof. Alberto Rodriguez.
- 2015–2016 **Massachusetts Institute of Technology (MIT)**, *Cambridge*, United States.
- Visiting student at the MCube Lab supervised by Prof. Alberto Rodriguez.
- 2011–2016 **Polytechnic University of Catalonia (UPC)**, *Barcelona*, Spain.
- Dual Bachelors in **Mathematics** and **Engineering Physics**.

Fellowships

- 2018 **Facebook Emerging Scholar Award**. Full funding for 2 years, only 21 awardees among more than 900 applications.
- 2018 **NVIDIA Graduate Fellowship**. Awarded to 10 PhD students from more than 230 applications. Declined in favour of Facebook Award.
- 2016 **"La Caixa" Graduate Fellowship**. Recipient of 1 out of the 45 prestigious "La Caixa" scholarships for graduate studies across all Spain. Full funding for 2 years in any graduate program of my choosing.
- 2012 **CFIS Fellowship**. Awarded to only 40 of the top technical students from Spain to simultaneously study two bachelor's degrees.

Selected Awards

- 2021 **Rising Stars in Computer Science.** Awarded to 89 graduate and postdoctoral women worldwide in EECS disciplines.
- 2021 **RSS pioneers.** Selected to attend the 2021 RSS Pioneers Workshop which brings together a cohort of the world's top early-career researchers in robotics.
- 2021 **Best Paper Finalist Award** on Service Robotics at ICRA 2021 for the work "Real-time shape and pose estimation from planar pushing using implicit surfaces."
- 2019 **Rising Stars in Mechanical Engineering.** Awarded to 30 graduate and postdoctoral women worldwide doing research in any area related to Mechanical Engineering.
- 2018 **Best Paper Award on Cognitive Robotics** at IROS 2018 for the work "Augmenting Physical Simulators with Stochastic Neural Networks: Case Study of Planar Pushing and Bouncing."
- 2018 Amazon Robotics **Best Systems Paper Award** for the submission "Robotic pick-and-place of novel objects in clutter with multi-affordance grasping and cross-domain image matching."
- 2017 **1st Place Winners** (Stow Task) at the international competition **Amazon Robotics Challenge** (ARC) 2017 with the MIT-Princeton team.
- 2016 **Best Paper Finalist Award** at IROS 2016 for the work "More than a Million Ways to Be Pushed. A High-Fidelity Experimental Data Set of Planar Pushing."

Publications

- [26] Z. Si, J. Chen, E. Karagozler, A. Bronars, J. Hutchinson, T. Lampe, N. Gileadi, T. Howell, S. Saliceti, L. Barczyk, I. Correa, T. Erez, M. Shridhar, M. Martins, K. Bousmalis, N. Heess, F. Nori, **M. Bauza**. "ExoStart: Efficient learning for dexterous manipulation with sensorized exoskeleton demonstrations", *submitted to ICRA 2026*.
- [25] **Gemini Team**. "Gemini 2.5: Pushing the frontier with advanced reasoning, multimodality, long context, and next generation agentic capabilities", *Technical Report*.
- [24] **Gemini Robotics Team**. "Gemini robotics: Bringing ai into the physical world", *Technical Report*.
- [23] A. Chen, P. Brakel, A. Bronars, A. Xie, S. Huang, O. Groth, **M. Bauza**, et al. "Exploiting Policy Idling for Dexterous Manipulation", *IROS 2025*.
- [22] **M. Bauza**, J. Chen, V. Dalibard, N. Gileadi, et al. "DemoStart: Demonstration-led auto-curriculum applied to sim-to-real with multi-fingered robots", *ICRA 2025*.
- [21] **M. Bauza**, T. Bronars, Y. Hou, I. Taylor, N. Chavan-Dafle, A. Rodriguez. "simPLE: a visuotactile method learned in simulation to precisely pick, localize, regrasp, and place objects", *Science Robotics 2024*.
- [20] Jacky Liang, Fei Xia, Wenhao Yu, Andy Zeng, Montserrat Gonzalez Arenas, Maria Attarian, **Maria Bauza**, et al. "Learning to learn faster from human feedback with language model predictive control", *RSS 2024*.
- [19] K. Bousmalis, G. Vezzani, D. Rao, C. Devin, A. Lee, **M. Bauza**, et al. "RoboCat: A Self-Improving Foundation Agent for Robotic Manipulation", *TMLR 2023*.
- [18] J. Zhao, **M. Bauza**, E. Adelson. "FingerSLAM: Closed-loop Unknown Object Localization and Reconstruction from Visuo-tactile Feedback", *ICRA 2023*.

- [17] **M. Bauza**, T. Bronars, A. Rodriguez. "Tac2Pose: Tactile Object Pose Estimation from the First Touch", *IJRR* 2023.
- [16] F. Alet, **M. Bauza**, K. Kawaguchi, N. Kuru, T. Lozano-Perez, L. Kaelbling. "Tailoring: Encoding Inductive Biases by Optimizing Unsupervised Objectives at Prediction Time", *NeurIPS* 2021.
- [15] S. Suresh, **M. Bauza**, A. Rodriguez, J. Mangelson, M. Kaess. "Real-time shape and pose estimation from planar pushing using implicit surfaces", *ICRA* 2021.
- [14] **M. Bauza**, E. Valls, B. Lim, T. Sechopoulos, A. Rodriguez. "Tactile Object Pose Estimation from the First Touch with Geometric Contact Rendering", *CoRL* 2020.
- [13] A. Kloss, **M. Bauza**, J. Wu, J. Tenenbaum, A. Rodriguez, J. Bohg. "Accurate Vision-based Manipulation through Contact Reasoning", *ICRA* 2020.
- [12] Y. Lin, **M. Bauza**, P. Isola. "Experience-Embedded Visual Foresight", *CoRL* 2019.
- [11] F. Alet, A. Jeewajee, **M. Bauza**, A. Rodriguez, T. Lozano-Perez, L. Kaelbling. "Graph Element Networks: adaptive, structured computation and memory", *ICML* 2019.
- [10] **M. Bauza**, O. Canal, A. Rodriguez. "Tactile Mapping and Localization from High-Resolution Tactile Imprints", *ICRA* 2019.
- [9] **M. Bauza**, F. Alet, Y. Lin, T. Lozano-Perez, L. Kaelbling, P. Isola, A. Rodriguez. "Omnipush: accurate, diverse, real-world dataset of pushing dynamics with RGB-D video", *IROS* 2019.
- [8] A. Ajay, **M. Bauza**, J. Wu, N. Fazeli, J. Tenenbaum, A. Rodriguez, L. Kaelbling. "Combining Physical Simulators and Object-Based Networks for Control", *ICRA* 2019.
- [7] **M. Bauza**, A. Rodriguez. "GP-SUM. Gaussian Processes Filtering of non-Gaussian Beliefs", *WAFR* 2018.
- [6] **M. Bauza**^{*}, F. Hogan^{*}, A. Rodriguez. "Learning vs. physics-based control of a planar push system", *CoRL* 2018.
- [5] **M. Bauza**^{*}, F. Hogan^{*}, A. Rodriguez. "Tactile Regrasp: Grasp Adjustments via Simulated Tactile Transformations", *IROS* 2018.
- [4] A. Ajay, J. Wu, N. Fazeli, **M. Bauza**, L. Kaelbling, J. Tenenbaum, A. Rodriguez. "Augmenting Physical Simulators with Stochastic Neural Networks: Case Study of Planar Pushing and Bouncing", *IROS* 2018.
- [3] A Zeng, S Song, K. Yu, E. Donlon, F. Hogan, **M. Bauza**, et al. "Active Perception of Novel Objects in Clutter with Multi-Affordance Grasping and Cross-Domain Image Matching" in *ICRA 2018, IJRR 2019*.
- [2] **M. Bauza**, A. Rodriguez. "A Probabilistic Data-Driven Model for Planar Pushing", in *ICRA 2017*.
- [1] K. Yu, **M. Bauza**, N. Fazeli, and A. Rodriguez. "More than a Million Ways to Be Pushed. A High-Fidelity Experimental Data Set of Planar Pushing," in *IROS 2016*.

Talks

- 2025 Invited talk at Deep Learning Barcelona Symposium.
- 2025 Invited talk at LaCaixa Becarios Knowledge Day.
- 2025 Invited talk at IROS Workshop Learning from Teleoperation.

2025 Invited talk at [IROS Workshop Tactile Sensing Toward Robot Dexterity and Intelligence.](#)

2025 Invited talk at [A Robot Touch of AI: London Summer School in Robotics & AI 2025.](#)

2025 Tech Talk at [ICRA Stage Presentations.](#)

2025 Invited talk at [ICRA Workshop Beyond Pick and Place.](#)

2025 Invited talk at [Microsoft Cortex AI Research Talk Series.](#)

2025 Invited guest lecture at [Oxford Saïd Business School MBA.](#)

2024 Invited talk at [CoRL Workshop on Learning Robotic Assembly.](#)

2024 Invited talk at [DevFest Menorca.](#)

2024 Invited talk at [Forum Illa del Rei on AI.](#)

2024 Invited talk at [Oxford Business school Analytics and AI class.](#)

2024 Invited talk at [ICRA Workshop Vitac.](#)

2024 Invited talk at ETH lab seminar.

2023 Invited talk at [Deep Learning Barcelona Symposium.](#)

2023 Invited talk and panel at [Balearic Ecosystem for AI.](#)

2023 Guest lecturer at Oxford's MBA class on Machine Learning for Business.

2023 Guest interview at [UK Robotics and Autonomous Systems Network's podcast, Robot Talk.](#)

2022 Invited talk at the RSS 2022 workshop on The Science of Bumping Into Things.

2022 Invited talk at [EPFL CS department](#)

2022 Invited talk at [Princeton ECE department](#)

2022 Invited talk at [CMU Robotics Institute](#)

2022 Invited talk at [University of Pennsylvania ECE department](#)

2022 Invited talk at [Columbia Mechanical Engineering department](#)

2022 Invited talk at [the Autonomy Talks at ETH Zurich](#)

2022 Invited talk at [Cornell ECE and CS departments](#)

2021 Invited talk at Washington University robotics colloquium.

2021 Invited talk at Stanford.

2021 Invited talk at CMU Manipulation Discussion Group (first external speaker).

2021 Invited talk at Cornell Robotic Seminar.

2021 Invited talk at University of Toronto, AI in Robotics Seminar.

2020 Invited talk at University of Pennsylvania, Grasp Seminar.

2020 Selected talk at SITN (Science In The News). Science talks organized by Harvard for the general public. Machine learning in Robotics: current progress and challenges ahead.

2019 Invited talk at Tec de Monterrey (Mexico).

2019 Selected presentation at the MIT College of Computing Launch.

2019 Selected talk at the ML across MIT retreat.

2018 Invited talk at IROS2018 Workshop on RoboTac: Tactile Perception and Learning in Robotics.

Advising

PhD and Master students

Joe Huang: Student researcher at GDM. Scaling tactile solutions.

Antonia Bronars: both at GDM and MIT. Implementation, development, and testing of tactile solutions for hands.

Bryan Lim: implementation of a grasping pipeline in simulation and on a real system.

Undergraduate students

Shreya Skarpoor: active 3D mesh reconstruction for object manipulation.

Claudia Lozano: application of machine learning methods to process tactile information.

Meenakshi Singh: simulation of a dual-arm in Pybullet with visuo-tactile sensing.

Eric Valls: implementation and development of tactile localization methods.

Max Thomsen: learning graph neural networks to optimize the shape of robotic fingers.

Oleguer Canals: implementation and testing of tactile algorithms for grasping and localization.

Jasmine Zeng: implementation of tactile sensing in a MuJoCo multi-fingered hand.

Theo Sechopoulos: implementation and comparison of registration techniques for tactile data.

Ashay Athalye: object tracking from RGB-D by extending a single-pose estimation algorithm.

Service

- 2022 Co-organized the workshop at ICRA 2022 Bi-manual Manipulation: Addressing Real-world Challenges.
- 2020, 2021 Guest lecturer at MIT graduate course *Touching and Grasping with Soft Fingers* led by Professor Ted Adelson.
- 2020 Program committee member of CoRL 2020.
- 2020 Co-organized the workshop at ICRA 2020 Uncertainty in Contact-Rich Interactions (canceled due to CoVID19).
- 2019 Organized and lead a hands-on robotic activity for the Women's Technology Program (WTP) in Mechanical Engineering.
- 2017 Co-organized the workshop at RSS 2017 Data-Driven Robotic Manipulation.
- 2016-present Journal reviewer: TRO, IJRR, RA-L.
- 2016-present Conference reviewer: RSS, CoRL, ICRA, IROS.
- 2011-2016 Class representative

Other Awards

- 2019 Selected to attend **Path of Professorship**. Awarded to distinguished MIT graduate students.
- 2019 Selected to attend the **Global Young Scientists Summit**. Awarded to no more than 5 PhD students throughout all MIT departments.
- 2019 **DeFlorez Travel Award in Design and Manufacturing**. Awarded to 1 MIT Mechanical Engineering Graduate Student annually.
- 2018 Travel Award to attend RSS 2018. The award is founded by the Women in Robotics Workshop.
- 2018 **Best Poster Award** at ICRA 2018 workshop on Active touch for perception and interaction.

- 2016 3rd Place at the 2016 Amazon Robotics Challenge with the MIT-Princeton team.
- 2016 IROS 2016 NSF Travel Grant.
- 2015 UPC-Internship Program Grant to do research for one year at MIT.
- 2015 Google Grace Hopper Travel Award to attend the conference with all expenses paid.
- 2011 6th position at the regional math tests: Kangourou sans frontieres.
- 2009 4th position at the regional math tests: Kangourou sans frontieres.

Press Coverage and Outreach

- 2024 The New Yorker. [A Revolution in How Robots Learn.](#)
- 2024 The Guardian. [Why aren't humanoids in our homes yet?.](#)
- 2024 MIT News Front page. [A new model offers robots precise pick-and-place solutions.](#)
- 2024 Diari Menorca interview. [Lo más importante es seguir conectados con el progreso.](#)
- 2023 Diari Mallorca interview. [Los robots harán los trabajos que no queremos para nuestros hijos.](#)
- 2022 Cornell News. [Creating Our Robotic Allies.](#)
- 2021 TechXplore. [A technique that allows robots to estimate the pose of objects by touching them.](#)
- 2021 La Vanguardia (Spanish newspaper). [Sensitive Robots.](#)
- 2019 MIT news article. [Pushy robots learn the fundamentals of object manipulation.](#)
- 2019 El Mundo (Spanish newspaper). [Una mano robotica inteligente.](#)
- 2019 El Iris (Spanish newspaper, front page). [Una ciutadellenca en el camp de la inteligencia artificial.](#)
- 2018 MIT News Front page. [Teaching robots how to move objects.](#)
- 2018 Invited talk for broad audience (100+ attendees). [First summer talk at Mercadal: Maria Bauza.](#)
- 2018 MIT MechEConnects. [Student Spotlight: Maria Bauza, PhD Candidate.](#)
- 2018 MIT News Front page. [Robo-picker grasps and packs.](#)
- 2018 Express News (Spanish version). [Maria Bauza, a Spanish woman who makes history.](#)
- 2018 La Vanguardia (Spanish newspaper). [MIT: the science paradise.](#)